Training and Inference Workflow of the Final Hierarchical System

Training path

Data protocol	Mechanism & XGBoost	Stage 1 detector	Size learning	Unified inverter
File1 train	Depth baseline bAcc 0.5138	AUC 0.8383	Size top1 0.6180	Hard-route depth bAcc 0.5337
File2 val	Spread / shape / deformation cues	Raw + delta	Size MAE 0.1565	GT / hard / soft / top2 route
File3 test	Provides interpretable reference	Detection-first design	Route supervision	Route-consistency regularization
Positive windows for inversion				

Inference path

Tactile sequence	Detection gate	Hierarchical inverter	Interpretability	Surgical UI
Raw amplitude	Always-visible probability	Size classification + regression	Latent probe	Probability curve
Normalized shape	Positive gate for inversion	Per-size depth experts	Branch ablation	Size / depth confidence
Delta dynamics		Confidence-aware route	Hard-pair / phase occlusion	Best-frame overview

Final system-level message

The final contribution is not a single best isolated metric. The system-level advance is that the hierarchical neural pipeline stably exceeds the structured XGBoost baseline on routed coarse depth under realistic deployment constraints while remaining interpretable.